FOURIER INTEGRAL & FOURIER TRANSFORM

$$f(x)=\frac{1}{\pi}\int_0^{\infty}[A(w)\cos wx+B(w)\sin wx]dx$$

$$A(w)=\int_{-\infty}^{\infty} f(x)\cos wx\,dx \qquad B(w)=\int_{-\infty}^{\infty} f(x)\sin wx\,dx$$

THE FOURIER INTEGRAL

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– Consider any periodic function of period $T=2L$.

$$f_L(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty} a_n\cos\omega_n x+b_n\sin\omega_n x \quad \text{——— } *$$

where $\omega_n=\frac{n\pi}{L}=\frac{2\pi n}{T}$ and $T=2L$.

What happens when $L\to\infty$? $\Sigma\to\int$ and $\omega_n\to\omega$, ω_n is no longer restricted to integer values

Put a_n & b_n in (*):

$$f_L(x)=\frac{2}{L}\int_{-L}^{L} f_L(v)\,dv+\frac{1}{L}\sum_{n=1}^{\infty}\left(\cos\omega_n x\int_{-L}^{L} f_L(v)\cos\omega_n v\,dv+\sin\omega_n v\int_{-L}^{L} f_L(v)\sin\omega_n v\,dv\right.$$

f_L is valid for arbitrarily large L, but fixed.

As $L \to \infty$, $f(x) = \lim_{L\to\infty} f_L(x)$ & $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$

$\frac{1}{L} \to 0$, $\Delta\omega = \frac{\pi}{L} \to 0$ and $\omega_n \longrightarrow \omega$.

Hence, f_L implies that: 28

$$f(x) = \frac{1}{\pi}\int_0^{\infty}\left[\cos\omega x \int_{-\infty}^{\infty} f(v)\cos\omega v\,dv + \sin\omega x \int_{-\infty}^{\infty} f(v)\sin\omega v\,dv\right] d\omega$$

Define: $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\,dv$ and $B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv$

$\therefore$ $$\boxed{f(x) = \int_0^{\infty}(A(\omega)\cos\omega x + B(\omega)\sin\omega x)\,d\omega} \quad \text{—, } \tilde{f}$$

This is the Fourier Integral.

where $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\,dv$ and

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv.$$

The Sufficient Conditions for the validity of $\mathcal{F}$, are as follows:

(1) – $f(x)$ is piecewise continuous in every finite interval.

(2) – $f(x)$ has a left-hand and a right-hand derivative at every point.

(3) – The integral $\int_{-\infty}^{\infty} |f(x)|\,dx$ exists i.e. absolutely integrable.

Then, $f(x)$ can be represented by $\mathcal{F}$. At a point of discontinuity the value of the Fourier Integral equals the average of the left-hand & right-hand limits of $f(x)$ at the point.

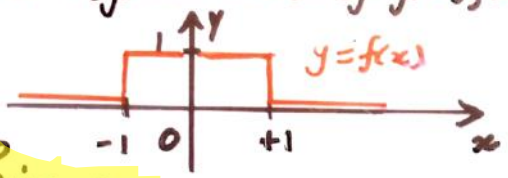
The Fourier Cosine Integral: $f(x) = \int_0^{\infty} A(\omega)\cos\omega x\,d\omega$, where

$$A(\omega) = \frac{2}{\pi}\int_0^{\infty} f(v)\cos\omega v\,dv.$$

The Fourier Sine Integral: $f(x) = \int_0^{\infty} B(\omega)\sin\omega x\,d\omega$, where

$$B(\omega) = \frac{2}{\pi}\int_0^{\infty} f(v)\sin\omega v\,dv.$$

30 Find the Fourier representation of $f(x)$, where $f(x) = \begin{cases} 1, & |x|<1 \\ 0, & |x|>1 \end{cases}$



Evaluate $\int_0^{\infty} \frac{\sin x}{x}\,dx$.

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\,dv = \frac{1}{\pi}\int_{-1}^{1}\cos\omega v\,dv = \frac{2\sin\omega}{\pi\omega}$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv = 0. \quad (\because f(v) \text{ is even.})$$

$x = 1$, a point of discontinuity: The average of left-hand right-hand limits at $x=1$: $\frac{0+1}{2} = \frac{1}{2}$.

right-hand limits at $x=1$: $\frac{0+1}{2} = \frac{1}{2}$.

$$\therefore f(x) = \int_0^\infty \frac{2\sin\omega}{\pi\omega}\cos\omega x\, d\omega = \begin{cases} 1, & 0 \le x < 1 \\ \frac{1}{2}, & x = 1. \\ 0, & x > 1. \end{cases}$$

This implies that :

$$\int_0^\infty \frac{\sin\omega}{\omega}\cos\omega x\, d\omega = \begin{cases} \pi/2, & 0 \le x < 1. \\ \pi/4, & x = 1. \\ 0, & x > 1. \end{cases}$$

Choose : $x = 0$

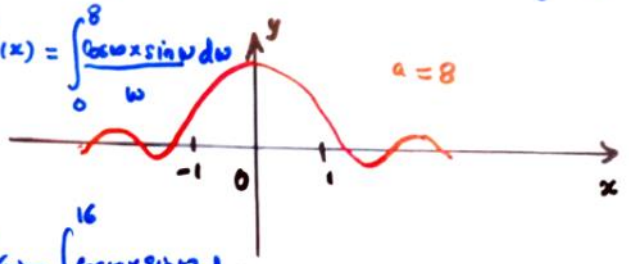
$$\int_0^\infty \frac{\sin\omega}{\omega}\, d\omega = \frac{\pi}{2},$$

$$\boxed{\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}.}$$

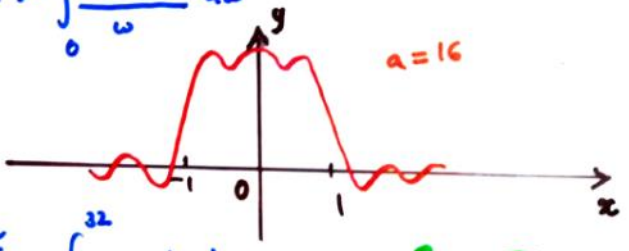
[Note: ω is a dummy variable, replace ω with x.]

$$f(x) = \frac{2}{\pi}\int_0^{\infty} \frac{\cos\omega x \sin\omega}{\omega}\, d\omega \approx \tilde{F}(x) = \frac{2}{\pi}\int_0^{a} \frac{\cos\omega x \sin\omega}{\omega}\, d\omega$$

$$y = \tilde{F}(x) = \int_0^{8} \frac{\cos\omega x \sin\omega}{\omega}\, d\omega$$

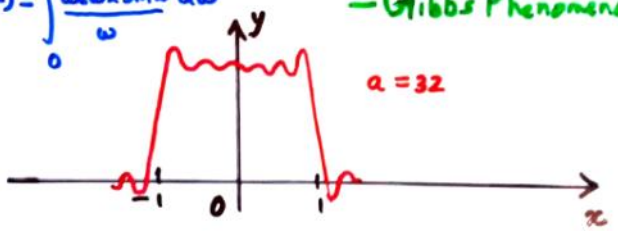


$$y = \tilde{F}(x) = \int_0^{16} \frac{\cos\omega x \sin\omega}{\omega}\, d\omega$$



$$y = \tilde{F}(x) = \int_0^{32} \frac{\cos\omega x \sin\omega}{\omega}\, d\omega$$

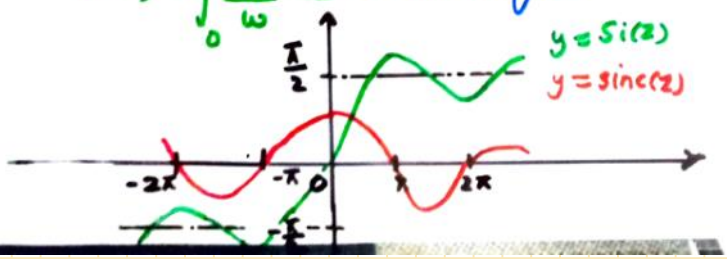
– Gibbs Phenomenon



Recall:

$$\operatorname{sinc}(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}, \quad \lim_{x\to 0} \frac{\sin x}{x} = 1.$$

$$Si(z) = \int_0^{z} \frac{\sin\omega}{\omega}\, d\omega \leftarrow \text{Sine Integral}$$



32/ Ex (1) Find the Fourier Cosine and Sine Integrals of $f(x)=e^{-kx}$, $x>0, k>0$.

Fourier Cosine Integral:

$$A(\omega)=\frac{2}{\pi}\int_0^{\infty} e^{-kv}\cos\omega v\,dv$$

By Parts Integration: $\int e^{-kv}\cos\omega v\,dv$

$$\int e^{-kv}\cos\omega v\,dv=\frac{-k}{k^2+\omega^2}e^{-kv}\left(-\frac{\omega}{k}\sin\omega v+\cos\omega v\right)+C$$

$$\therefore \int_0^{\infty} e^{-kv}\cos\omega v\,dv = 0+\frac{k}{k^2+\omega^2}$$

$$\therefore A(\omega)=\frac{2k}{\pi(k^2+\omega^2)}$$

The Fourier Cosine Integral Representation:

$$\therefore f(x)=\int_0^{\infty} A(\omega)\cos\omega x\,d\omega$$

$$=\int_0^{\infty}\frac{2k}{\pi(k^2+\omega^2)}\cos\omega x\,d\omega$$

$$=\frac{2k}{\pi}\int_0^{\infty}\frac{\cos\omega x}{k^2+\omega^2}\,d\omega = e^{-kx}, \quad x>0, k>0.$$

Hence,

$$\int_0^{\infty}\frac{\cos\omega x}{k^2+\omega^2}\,d\omega=\frac{\pi}{2k}e^{-kx}, \quad x>0, k>0. \dagger$$

†. This integral is called a Laplace integral.

Fourier Sine Integral:

$$B(\omega) = \frac{2}{\pi}\int_0^\infty f(v)\sin\omega v\,dv$$

$$= \frac{2}{\pi}\int_0^\infty e^{-kv}\sin\omega v\,dv$$

By integration by parts: $\int e^{kv}\sin\omega v\,dv$

$$\int e^{-kv}\sin\omega v\,dv = \frac{-\omega}{k^2+\omega^2}e^{-kv}\left(\frac{k}{\omega}\sin\omega v + \cos\omega v\right)$$

$$\therefore \int_0^\infty e^{-kv}\sin\omega v\,dv = \frac{\omega}{k^2+\omega^2}.$$

$$\therefore B(\omega) = \frac{2\omega}{\pi(k^2+\omega^2)}$$

The Fourier Sine Integral Representation:

$$f(x) = \int_0^\infty B(\omega)\sin\omega x\,dx$$

$$= \frac{2}{\pi}\int_0^\infty \frac{\omega\sin\omega x}{k^2+\omega^2}\,d\omega = e^{-kx}, \quad k>0, x>0.$$

Hence, $$\int_0^\infty \frac{\omega\sin\omega x}{k^2+\omega^2}\,d\omega = \frac{\pi}{2}e^{-kx}, \quad k>0, x>0.$$ †

†: This integral is called a Laplace integral.

Fourier Integral Representations:

2) $$\int_0^\infty \frac{\cos wx + w\sin wx}{1+w^2}\,dw = \begin{cases} 0, & \text{if } x<0 \\ \frac{\pi}{2}, & \text{if } x=0 \\ \pi e^{-x}, & \text{if } x>0. \end{cases}$$

3) $$\int_0^\infty \frac{w^3 \sin xw}{w^4+4}\,dw = \frac{\pi}{2} e^{-x}\cos x \quad \text{if } x>0.$$

4) $$\int_0^\infty \frac{\cos xw}{1+w^2}\,dw = \frac{\pi}{2} e^{-x} \quad \text{if } x>0.$$

5) $$\int_0^\infty \frac{\sin w \cos wx}{w}\,dw = \begin{cases} \pi/2, & 0 \le x < 1. \\ \pi/4, & x = 1. \\ 0, & x > 1. \end{cases}$$

6) $$\int_0^\infty \frac{\sin \pi w \sin xw}{1+w^2}\,dw = \begin{cases} \frac{\pi}{2}\sin x, & 0 \le x \le \pi \\ 0, & x > \pi. \end{cases}$$

Exercises:

Recall: $\int e^{ax}\sin bx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$.

$$\int e^{ax}\cos bx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C.$$

2) $$\int_0^\infty \frac{\cos\omega x + \omega\sin\omega x}{1+\omega^2}\,d\omega = \begin{cases} 0, & x<0. \\ \pi/2, & x=0. \\ \pi e^{-x}, & x>0. \end{cases}$$

We know that $f(x) = \int_0^\infty (A(\omega)\cos\omega x + B(\omega)\sin\omega x)\,d\omega$.

where $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\,dv$ and $B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv$.

Take: $f(x) = \begin{cases} e^{-x}, & x>0. \\ 0, & x<0. \end{cases}$

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\,dv = \frac{1}{\pi}\int_0^\infty e^{-v}\cos\omega v\,dv$$

$$= \frac{1}{\pi}\left[\frac{e^{-v}}{1+\omega^2}(-\cos\omega v + \omega\sin\omega v)\right]_0^\infty$$

$$= \frac{1}{\pi}\left\{0 + \frac{1}{1+\omega^2}\right\} = \frac{1}{\pi(1+\omega^2)}.$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\,dv = \frac{1}{\pi}\int_0^\infty e^{-v}\sin\omega v\,dv$$

$$= \frac{1}{\pi}\left[\frac{e^{-v}}{1+\omega^2}(-\sin\omega v - \omega\cos\omega v\right]_0^\infty$$

$$= \frac{1}{\pi}\left\{0 + \frac{\omega}{1+\omega^2}\right\} = \frac{\omega}{\pi(1+\omega^2)}.$$

$$*,\quad f(x)=\int_0^\infty\left(\frac{1}{\pi(1+\omega^2)}\cos\omega x+\frac{\omega}{\pi(1+\omega^2)}\sin\omega x\right)d\omega$$

$$=\frac{1}{\pi}\int_0^\infty\frac{\cos\omega x+\omega\sin\omega x}{1+\omega^2}\,d\omega=\begin{cases}0, & x<0\\ \frac{1}{2}, & x=0\\ e^{-x}, & x>0\end{cases}.$$

$$*,\Rightarrow\int_0^\infty\frac{\cos\omega x+\omega\sin\omega x}{1+\omega^2}\,d\omega=\begin{cases}0, & x<0\\ \frac{\pi}{2}, & x=0\\ \pi e^{-x}, & x>0\end{cases}.$$

Note: f is discontinuous at $x=0$,
so the integral converges to the average value $\frac{0+1}{2}=\frac{1}{2}$.